Kelechi Emeka Ogbonna

Computational modelling & simulation · biology, autonomy and spaceflight systems

Nigeria · kelechiogbonna300@gmail.com · github.com/cloudynirvana

PROFILE

Biotechnology graduate who builds tested, reproducible simulation software: dynamical-systems models of disease and physiology, agent-based models of decision-making under uncertainty, and reinforcement-learning control. I report what results do and do not show, including negative findings, and I back claims with held-out evaluation and proper statistics. Seeking internships or graduate roles in computational modelling, space life sciences, or simulation and autonomy.

SELECTED PROJECTS

Adaptive Convergence: simulating R&D strategy for a spacefaring civilisation 2026

- Built an agent-based simulator of long-horizon technology development (procedural technology graphs with competing routes, hidden dead ends, Bayesian cost learning, recursive budget growth) and benchmarked six strategies against a perfect-information oracle.
- Committing to the best option while abandoning projects on statistical evidence of overrun gave **+15.8% capability** over the best convergent baseline (95% CI 13.0–18.9%, better in 75% of 400 held-out worlds). It **cut budget lost to dead ends from 21% to 9%** and raised interstellar-milestone success from 41% to 56%.
- A 20-condition sweep showed the advantage exists only when failures lie outside the planner's model; ablations showed extra exploration *reduced* performance. Found and fixed a belief-model bug, verified the fix against Monte Carlo, and reran every experiment.
- Replicated March (1991, *Organization Science*). Held-out tuning seeds, bootstrap CIs, Wilcoxon tests with Holm correction, 21 unit tests.

Python · NumPy · SciPy · Matplotlib · pytest

Rocket Lander RL: reward shaping, reward hacking and robustness 2026

- Trained PPO agents to land a rocket (Gymnasium LunarLander) under four reward designs × 5 seeds. Evaluated every agent on the unshaped reward so shaping cannot inflate scores.
- Reproduced and measured a reward hack (agents hovering to farm a per-frame bonus), then removed it with potential-based shaping (Ng et al., 1999) plus a time cost.
- Monte-Carlo-tested agents and a hand-written control law under wind and gravity dispersions.

Python · PyTorch · Stable-Baselines3 · Gymnasium · pytest

Project Confluence: dynamical models of cancer metabolism *YYYY–YYYY*

- ODE framework for metabolic reprogramming across cancer types, with structural identifiability analysis and adaptive-control experiments.
- Fitting to CCLE metabolomics (928 cell lines) raised identifiable parameters from 7 to 15 of 17. About 190 Python modules and 330+ automated tests in GitHub Actions. All results are in silico.

Python · SciPy · ODEs · identifiability · pytest · GitHub Actions · FastAPI

TNBC metabolic strain model (M.Sc. research) *YYYY*

- ODE and Monte Carlo model of ATP, reactive-oxygen-species and glucose dynamics in triple-negative breast cancer, with stability and bifurcation analysis and parameter sweeps.

EDUCATION

M.Sc. Biotechnology · Ahmadu Bello University, Zaria In progress
Systems biology, nonlinear dynamics, bioinformatics, bioprocess engineering.

Postgraduate Diploma, Aviation Management · Nigerian College of Aviation Technology, Zaria In progress

B.Sc. Biotechnology (Hons) · Nile University of Nigeria, Abuja 2022
Project: green synthesis of silver nanoparticles from *Carica papaya* leaf extract; antioxidant and oxidative-stress assays.

EXPERIENCE

Biotechnology Instructor · NYSC, Etinan College of Health and Technology, Akwa Ibom *YYYY–YYYY*

- Designed and taught biotechnology modules, including introductory mathematical modelling of biological processes; led laboratory sessions and supervised student projects with limited equipment.

Research Intern · National Biotechnology Development Agency (NABDA), Abuja *YYYY*

- Molecular sequencing workflows and laboratory analysis on genetically modified crop projects; exposure to biosafety regulation.

Laboratory Intern · Arubah Medical Diagnostic Services, Calabar *YYYY*

- Processed clinical samples by PCR and spectrophotometry under laboratory quality-control procedures.

Managing Director · Uloma Shopping Mall *YYYY–YYYY*

- Ran day-to-day operations, finances, inventory and supplier relationships for a retail business. *Add one number: staff, revenue, or an improvement you made.*

POS Operator · MoneyPoint Microfinance Bank *YYYY–YYYY*

- Processed electronic payments accurately and in line with banking compliance procedures.

SKILLS

Programming	Python (NumPy, SciPy, Matplotlib, PyTorch, Stable-Baselines3, Gymnasium, pytest), MATLAB, Git and GitHub Actions, Jupyter/Colab
Modelling	ODE modelling, stability and bifurcation analysis, structural identifiability, Monte Carlo, agent-based simulation, Bayesian updating, reinforcement learning
Statistics	Bootstrap confidence intervals, nonparametric and paired tests, multiple-comparison correction, experimental design with held-out evaluation
Laboratory	PCR, DNA sequencing workflows, spectrophotometry, nanoparticle synthesis, enzyme and antioxidant assays, BLAST

CERTIFICATIONS

IELTS General 7.5 · Google Project Management (Coursera) · Anthropic Academy: AI Fluency, Claude 101, Model Context Protocol (intro and advanced)